Factory Reallocation & Shipping Optimization Recommendation System

Nassau Candy Distributor

Project Report

Abstract

In this project, I developed a factory reallocation and shipping optimization system for Nassau Candy Distributor. The main idea was to check whether changing the factory assigned to a product could improve shipping lead time. I used the available order data for data cleaning, feature engineering, machine learning, route analysis and scenario testing.

I tested three models: Linear Regression, Random Forest and Gradient Boosting. Random Forest gave the best result among the three models and was used in the scenario engine. I then created a system that compares the current factory with the other available factories and gives a recommendation based on predicted lead time. Finally, I connected the analysis to a Streamlit dashboard so that the results can be viewed using simple selections such as product, region and shipping mode.

1. Introduction

The purpose of this project is not only to understand what happened in the shipping data, but also to help answer a practical question: what could be changed to improve the result?

Nassau Candy has different products assigned to different factories. If a product is shipped from a factory that is not suitable for a particular destination, the shipping time can become higher. Instead of changing assignments manually, I wanted to build a small decision-support system that can compare different factory choices before making a change.

The project therefore combines normal data analysis with machine learning and what-if scenario analysis.

2. Problem Statement

The existing factory assignments are based on fixed or legacy processes. This can lead to longer shipping times for some product and region combinations. There is also no simple way to test an alternative factory and see the possible effect before actually changing the assignment.

The project aims to solve this by predicting lead time for different factory scenarios and giving a recommendation when another factory is predicted to perform better.

3. Objectives

- Clean and prepare the Nassau Candy dataset.

- Create useful features for machine learning.
- Predict shipping lead time.
- Compare Linear Regression, Random Forest and Gradient Boosting.
- Study route performance using clustering.
- Simulate different factory assignments.
- Calculate predicted lead-time reduction.
- Show confidence and risk information.
- Build a simple Streamlit dashboard for the final results.

4. Dataset

The dataset contains order, customer, product, sales, cost, profit and shipping information. Some of the important fields used in the project are shown below.

Field	Meaning
Order Date	Date when the order was placed
Ship Date	Date when the order was shipped
Ship Mode	Shipping method
Region	Customer region
Division	Product division
Product Name	Name of the product
Sales	Sales value
Units	Number of units
Gross Profit	Sales minus cost
Cost	Manufacturing cost

5. Factory Information

The project uses the five factories supplied in the project requirements:

Factory	Latitude	Longitude
Lot's O' Nuts	32.881893	-111.768036
Wicked Choccy's	32.076176	-81.088371
Sugar Shack	48.11914	-96.18115
Secret Factory	41.446333	-90.565487
The Other Factory	35.1175	-89.971107

6. Methodology

6.1 Data Preparation and Feature Engineering

I first prepared the data so it could be used by the models. Date information was split into useful parts such as Order Year, Order Month, Order Day and Order Day of Week. Product, factory, region, ship mode and division information were also included in the feature dataset.

The cleaned and engineered data was saved separately so that the same prepared data could be reused in the later stages of the project.

6.2 Machine Learning Models

I tested three regression models for predicting shipping lead time:

- Linear Regression was used as the baseline.
- Random Forest Regressor was tested for non-linear relationships.
- Gradient Boosting Regressor was also tested as another non-linear model.

6.3 Model Evaluation

The models were compared using MAE, RMSE and R^2. MAE shows the average absolute prediction error. RMSE gives more importance to larger errors. R^2 shows how much of the variation in the target is explained by the model.

The actual results from my model run were:

Model	MAE	RMSE	R^2
Linear Regression	181.422	182.490	0.5291
Random Forest	135.405	160.466	0.6359
Gradient Boosting	159.638	169.039	0.5960

Random Forest was selected because it had the lowest RMSE. It also had the lowest MAE and the highest R^2 in this comparison. The trained model was saved as best_lead_time_model.pkl.

6.4 Route Clustering

I also added a clustering stage to look at route performance. The purpose was to group similar route combinations and identify slower combinations that may need more attention. The project generates clustered data, route performance data and a cluster summary.

6.5 Factory Scenario Simulation

The scenario engine is the main decision-support part of the project. The user selects a product, region and ship mode. The program finds the current factory and creates five scenarios, one for each factory.

For the selected combination, typical numerical values are taken from the available historical data. If an exact product-region-ship-mode combination is not available, the program uses product-level data instead. The trained model then predicts the lead time for each factory scenario.

6.6 Recommendation Logic

The current factory prediction is used as the reference. The predicted lead time for every alternative factory is compared with it. If an alternative has a lower predicted lead time, the system can recommend that factory. If none of the alternatives is better, the result is KEEP CURRENT FACTORY.

This avoids forcing a recommendation when the model does not show an improvement.

6.7 Profit, Confidence and Risk

The dashboard also shows profit margin, sample count, confidence score and risk level. The confidence score is based on the number of historical observations available for the selected scenario and is capped at 100%.

For profit impact, I did not make up factory-specific costs. The current dataset does not contain manufacturing costs for hypothetical reassignment to another factory. Because of this, the dashboard clearly shows that hypothetical profit impact is treated as stable. This is a limitation of the current version.

7. Streamlit Application

After completing the analysis, I created a Streamlit application to make the project easier to use. Instead of running the scenario engine manually in the terminal, the user can select values from the sidebar and view the results on the page.

- Product selector
- Region selector
- Ship mode selector
- Speed versus profit priority slider
- Factory Optimization Simulator
- Risk & Impact Panel
- Profit margin and sample count
- Decision summary and recommendation

8. Results

The machine learning comparison showed that Random Forest performed best among the tested models. Its RMSE was 160.466, compared with 182.490 for Linear Regression and 169.039 for Gradient Boosting.

The scenario engine successfully compared the five factories. In the demonstrated runs, the system could return KEEP CURRENT FACTORY when no alternative had a lower predicted lead time. This is useful because the system is designed to make a recommendation only when the model shows an improvement.

9. Key Performance Indicators

KPI	Used in the project
Lead Time Reduction (%)	Difference between current and scenario predicted lead time as a percentage of current prediction.
Profit Impact Stability	Shown as stable because factory-specific hypothetical costs are not available.
Scenario Confidence Score	Based on the number of matching historical observations.
Recommendation Coverage	The engine checks all five supplied factory options.

10. Business Recommendations

- Use the scenario engine before changing a product's factory assignment.

- Give more attention to scenarios with large predicted lead-time improvements.
- Check the confidence and risk information before acting on a recommendation.
- Keep the current factory when no alternative is predicted to improve lead time.
- Collect factory-specific manufacturing and transportation costs for better profit analysis.
- Use the route analysis to identify recurring slow routes.

11. Limitations

- The Random Forest R^2 of 0.6359 shows that there is still variation in lead time that the model does not explain.
- The current dataset does not contain factory-specific costs for hypothetical reassignment.
- Product-level data is used as a fallback when an exact scenario has no historical observations.
- The confidence score is a simple project-level measure based on sample count; it is not a statistical probability.
- The recommendations should be checked against real factory capacity, inventory and operational constraints before being used in production.

12. Future Scope

- Add actual factory-specific production and shipping costs.
- Use factory coordinates to add distance-related features.
- Add factory capacity and inventory constraints.
- Add better uncertainty or prediction-interval estimates.
- Retrain the model regularly when new order data is available.
- Improve the dashboard with historical trend charts and downloadable reports.

13. Conclusion

I developed a working prototype that connects data analysis, machine learning and factory scenario testing for Nassau Candy Distributor. The project does not stop at predicting shipping lead time; it uses the prediction to compare different factory choices and provide a simple recommendation.

Among the three tested models, Random Forest performed best with an MAE of 135.405, RMSE of 160.466 and R^2 of 0.6359. The Streamlit dashboard makes the scenario engine easier to use and also shows the limitations of the current data. Overall, the project provides a practical starting point for improving factory assignment decisions, with actual cost and operational data being the main requirements for a more complete optimization system.